

Develop a program using malloc(), calloc(), realloc(), and free(). Monitor memory allocation behavior and identify memory leaks using Valgrind.

Develop a program that demonstrates Copy-on-Write behavior after a fork(). Analyze memory usage before and after modifying data in the child process.

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Last login: Sat Sep 19 13:36:52 on ttys001
[varnikakomali@VARNIKAs-MacBook-Air ~ % cd OS_Lab
[varnikakomali@VARNIKAs-MacBook-Air OS_Lab % cd Practical
[varnikakomali@VARNIKAs-MacBook-Air Practical % nano practical8.c
[varnikakomali@VARNIKAs-MacBook-Air Practical % gcc practical8.c -o practical8 && ./practical8
=== Part 1: malloc, calloc, realloc, free & Leaks ===
malloc array: 10 20 30
calloc zeroed: 0 0 0
realloc array: 10 20 30 40 50
Freed allocated buffers successfully.
Simulated memory leak: 256 bytes allocated without free().

=== Part 2: Copy-on-Write (COW) Behavior ===
Parent allocated 1 MB buffer initialized to 'A'.
Parent data address: 0x734c00000, sample char: 'A'

[Child] Before modification (Shared physical frame via COW):
[Child] Address: 0x734c00000, sample char: 'A'
[Child] Modifying buffer...
[Child] After modification (COW duplicate frame allocated):
[Child] Address: 0x734c00000, sample char: 'B'

[Parent] Child finished.
[Parent] Value remains unchanged: 'A'
varnikakomali@VARNIKAs-MacBook-Air Practical % █
```